

Engineering 10: Applied Engineering Computer Methods Spring 2023 Course Syllabus (Subject to Change)

Professor	Prof. George Witmer (gsw2@lehigh.edu)
Class Meetings	January 23 – May 3: Monday-Wednesday 7:55 AM, Packard Lab 101 Class attendance requirements based on Group (see below) Lab: <u>Fritz 402</u> , attendance mandatory. Please check your lab section. Students are required to bring their laptops to class and lab.
Office Hours	Monday, Wednesday 11:00-1:00 Office: Packard Lab 206A
Piazza	TBD
Textbook/Resources	Learn Python in One Day and Learn It Well , Jamie Chan. \$2.99 eBook No textbook for Robotics portion.
Course Prerequisites	Students should be comfortable with basic scientific principles as taught in high school chemistry and physics classes, as well as math through trigonometry. No prior experience with computers is required.

Academic Honesty

All work is to be completed on your own. Student submissions can easily be compared one to another. Plagiarism will not be tolerated. (<https://libraryguides.lehigh.edu/plagiarism>).

All students caught cheating will receive a zero for that assignment. Moreover, they will be reported to the Office of Student Conduct. This will result in a *permanent record* being placed in their file.

Cheating includes but is not limited to the following:

- 1) **Uploading, downloading or viewing answers to/from internet sites, such as Chegg.com, Course Hero, etc.**
- 2) **Code generators such as ChatGPT**
- 3) **Providing, receiving or viewing code from current or past students of this course**
- 4) **Viewing answer keys in any way possible**

Course Description: Introduction to programming for engineering tasks. Use of Python to program and solve engineering problems. Interfacing sensors and actuators to a microcontroller board and programming to interact with the world.

Course Format: In this course, you will learn the basics of programming using Python and use it to perform computations and perform data analysis. You will also apply programming to monitor and control hardware using the Raspberry Pi and a robot kit.

Inclement Weather Policy: In case of inclement weather, I will hold class on Zoom. I will send an email with the Zoom link ahead of class via Course Site, and will also post the Zoom link on the Course Site. If the University closes down, your attendance is not required. Otherwise, I will take attendance as usual. The class recording will be posted on the Course Site.

Course Learning Objectives and Outcomes: By the end of this course, you'll be able to:

- Develop and run software applications using standard programming constructs such as loops, arrays, functions, and conditional statements
- Use standard scientific libraries available for Python, including NumPy, SciPy, Pandas and Matplotlib to solve engineering problems and analyze data
- Input and output data from/to various devices, including keyboard, files, and external sensors and devices
- Control and monitor a variety of hardware systems composed of multiple electronic components

We also expect the course to:

- Teach general approaches to solving engineering problems
- Introduce a variety of engineering disciplines via lectures and exercises
- Bring an awareness to students of some of the ethical challenges posed by the evolution of technology in society.

Course Grading Percentage Scales (Less than 60% is a failing grade):

D	D+	C-	C	C+	B-	B	B+	A-	A
60-63	67-70	70-73	73-77	77-80	80-83	83-87	87-90	90-93	93-100

Grades are not rounded up or down!

Learning Groups: Because students come to this class with different levels of programming experience, three learning Groups are available for the Python lectures and assignments. Group 1 is for students with little or no programming experience. Group 2 is for students who have learned to program in a different language, typically Java. Group 3 is for students who already program in Python. Students select their own Group at the beginning of the semester; changing Groups requires instructor approval. Requirements for each Group are as follows:

Group ->	1	2	3
Python lecture attendance	Monday lecture required Wednesday lecture optional	Monday lecture optional Wednesday Lecture required	Python Lectures optional
Robotics lecture attendance	Monday lecture required. Wednesday help section optional	Monday lecture required. Wednesday help section optional	Monday lecture required, Wednesday help section optional
Lab Attendance	Must attend lab every week (must stay for the entire lab)	Must check in to lab every week (can leave after check-in)	Attendance required only for Robotics assignments
Quizzes	Yes	No	No
Python Assignments	Python Group 1 (weekly)	Python Group 2 (weekly)	Python Group 3 (weekly)

Engineering Assignments	Group 1	Group 1 + 2	Group 1 + 2 + 3
Lab Assignments	Required	Required	Required

Assignments:

- Weekly Quizzes (12 quizzes)
 - Quizzes will be provided on Course Site for **Group 1 students only**
 - Quizzes will be based on lecture material covered during class or previous classes
 - Quizzes will cover general engineering concepts and basic Python syntax
 - Quizzes are open book, open notes and are timed (30 minute limit)
- Python Assignments (7 assignments)
 - Python assignments are intended to provide students practice using the concepts taught in the lectures
 - Assignments for Group 1 will include more numerous, basic problems, Group 2 will have fewer problems and will focus on syntax unique to Python, and Group 3 will have the fewest, but most challenging, problems.
 - Python assignments will be submitted to Course Site
- Engineering problem assignments (8 Assignments)
 - Engineering assignments require you to solve an engineering problem by writing a Python program
 - Your grade will be based primarily on your written description of your project approach and reasoning with the remainder based on your method and results (see discussion of rubric below).
 - To ensure your work is graded fairly, you may be asked to provide additional information about your assignment, in a written or oral format
 - The assignments have sections for Groups 1, 2 and 3. Group 1 will do section 1 only. Group 2 will do sections 1 and 2; Group 3 will do sections 1, 2 and 3. **For your convenience, the Group 2 and 3 assignments are to be submitted in a single notebook that includes all required sections.**
- Lab Assignments (6 assignments)
 - Lab assignments give you the opportunity to do hands-on manipulation of circuits and electronic components as well as to control them using Python
 - Students are required to demonstrate their completed work to a TA in the lab
 - Students will also submit their working Python files to Course Site
 - Each lab outlines requirements for a C, B or A grade.
 - Groups 1, 2, 3 all do the same lab assignments

Late Assignments: Because the deadlines are quite lenient and ample time has been provided for the assignments, **no extensions will be granted, and no credit given for late assignments.**

- No extensions because you were ill (unless your doctor provides a note indicating you were ill for **a week or more**)
- No extensions because your computer broke down (use a public computer and your data which was stored on a cloud drive, or borrow a laptop from LTS)
- No extensions due to travel for sports, conferences, family events/emergencies, etc. (plan ahead and use your time wisely)
- No extensions because “it’s a really busy week and I have several exams” (don’t even ask)

To ensure that you finish your assignments on time, we encourage you to start and finish early. It is your responsibility to see that your assignments are correctly submitted on time. If you have any questions, please contact your TA **before the assignment is due**.

Quizzes: The purpose of these is to aid in your retention. Scientific studies of learning have demonstrated that being required to recall material leads to better long-term retention. The quizzes are required for Group 1 students only and should be completed before attempting the other assignments.

Python Assignments: These will be straightforward applications of concepts taught in class. The purpose of these assignments is to help you develop proficiency in programming. Whereas the quizzes will assist you to **learn** Python, the assignments will develop your ability to **program** in Python. These assignments will be given credit based on completion; if all parts of the assignment are completed, full credit will be given (A, 95%).

Engineering Problem Assignments: These assignments provide an opportunity to **apply** what you have learned by solving a practical engineering problem. Group 1 students are required to complete the Group 1 section; Group 2 students must also complete the group 2 section, and Group 3 students must complete all 3 sections (1, 2 and 3) of the engineering assignment. Full credit (A, 95%) will be given for each section for completion of all requirements of that section. Exceptional work that goes beyond the assignment requirements may receive up to 100%.

Lab Assignments: These assignments provide an opportunity for you to **apply** what you have learned by building and controlling a physical device. They must be completed in the lab. Full credit (A, 95%) is given for completion of all requirements. Partial completion will result in a B (85%) or C (75%). As with all assignments, lab assignments must be completed on time, and no extensions or late credit will be given.

Rubric for Engineering Assignments: Engineering assignments are to be submitted on CourseSite and will be graded according to a rubric. The general guidelines of the rubric are as follows:

50%	A clear written explanation, demonstrating a full understanding of the problem, the solution approach, and the results
30%	Correctness of the approach. Were all requirements met? Were all factors considered? Did you take advantage of the available programming tools and features?
20%	Correctness of the results. Reasonable results receive most of the credit; unreasonable results, i.e. those that defy common experience, receive little or no credit.
(-20%)	Instructions must be followed: correct files submitted; assignment formatted correctly - or else up to 20% may be deducted from your score

You Learn by Doing Not Watching: Learning to program and learning to solve engineering problems are much like swimming or riding a bicycle. You cannot learn it just by being told how to do it, by watching someone else do it, or by copying someone else's work. It is critical that you completely carry out each assignment on your own. Some struggling and frustration is a normal and even essential part of the learning process. However, if you are struggling and still failing to understand, seek assistance! Please contact your TA's or the professors promptly – don't wait until late in the course. Concepts presented later in the course will assume you have mastered earlier concepts, and if this is not the case, you may not be able to keep up with the class.

Piazza Workspace: We will be using Piazza for class discussion. Piazza is designed to get you help fast and efficiently from classmates, the TAs, and me. Students are instructed to post questions, comments, and additional solutions on Piazza, where the instructor and TAs are available to help. **No assistance will be**

provided via email. Use Piazza or arrange to stop by during office hours. You can post a DM to the professor on Piazza with questions about grades, etc.

Our class page:: piazza.com/lehigh/fall2022/engr10/home

Lab Attendance: The course includes lab sessions hosted by a Teaching Assistant, who is there to answer questions to help your understanding and to check you off on your robotics assignments. Group 1 **must** sign in with their TA at the lab at least once per week. All groups must attend the lab to complete their lab assignments.

Class Recordings: The weekly class lecture will be recorded via Zoom. Permission can be granted to students who need to attend class remotely via the Zoom link by exception due to illness. If you feel you have a valid case, please contact the professor at least 30 minutes before class. Class recordings will be posted on the CourseSite and you are encouraged to review them as needed.

Students must bring their laptops to class. The lectures will use an interactive environment called Jupyter Notebook. Students are invited to download the lectures in advance and follow along during class. Students will also use their laptops for in-class surveys and polls, as well as to take attendance.

Grading

Group ->	1	2	3
Attendance	5%	5%	0%
Quizzes	10%	0%	0%
Python Assignments	20%	20%	10%
Engineering Assignments	20%	30%	45%
Lab Assignments	35%	35%	35%
Final Lab	10%	10%	10%
Total	100%	100%	100%

Accommodations for Students with Disabilities

Lehigh University is committed to maintaining an equitable and inclusive community and welcomes students with disabilities into all of the University's educational programs. In order to receive consideration for reasonable accommodations, a student with a disability must contact Disability Support Services (DSS), provide documentation, and participate in an interactive review process. If the documentation supports a request for reasonable accommodations, DSS will provide students with a Letter of Accommodations. Students who are approved for accommodations at Lehigh should share this letter and discuss their accommodations and learning needs with instructors as early in the semester as possible. For more information or to request services, please contact Disability Support Services in person in Williams Hall, Suite 301, via phone at 610-758-4152, via email at indss@lehigh.edu, or online at <https://studentaffairs.lehigh.edu/disabilities>.

Lehigh University Policy on Harassment and Non-Discrimination

Lehigh University upholds The Principles of Our Equitable Community and is committed to providing an educational, working, co-curricular, social, and living environment for all students, staff, faculty, trustees, contract workers, and visitors that is free from harassment and discrimination on the basis of age, color, disability, gender identity or expression, genetic information, marital or familial status, national or ethnic origin, race, religion, sex, sexual orientation, or veteran status. Such harassment or discrimination is unacceptable behavior and will not be tolerated. The University strongly encourages (and, depending upon the circumstances, may require) students, faculty, staff or visitors who experience or witness harassment or discrimination, or have information about harassment or discrimination in University programs or activities, to immediately report such conduct.

If you have questions about Lehigh's Policy on Harassment and Non-Discrimination or need to report harassment or discrimination, contact the Equal Opportunity Compliance Coordinator (Alumni Memorial Building / 610.758.3535 / eocc@lehigh.edu)